Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	David Hu	**Email**	hu@me.gatech.edu
Institute Name	Georgia Institute of Technology			
Address	801 Ferst Dr NW, Atlanta, GA 30318			

2. Technology Overview

Title of Proposed Technology	Development of Governing Equations and Multiscale Simulation Methodology for Nature-Inspired Porous Media in High-Density Data Center Cooling			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Machinery and Materials			
	Medium Category	Small Category	Medium Category	Small Category
	General Machinery / Fluid Machinery	High-efficiency Heat Exchangers		
Concept of Proposed Technology	1. Problem Definition (The Complexity Challenge): Nature-inspired repetitive 3D structures offer superior thermal density compared to traditional 2D plate heat exchangers. However, the complex internal flow fields create chaotic turbulence and boundary layer interactions that are computationally prohibitive to simulate at a full scale using Direct Numerical Simulation (DNS). 2. Theoretical Approach (Deriving Governing Equations): Instead of modeling every geometric detail, we treat the structure as a "Porous Medium." By analyzing the micro-scale flow physics within a representative cell, we derive new			

	Volume-Averaged Governing Equations. These equations mathematically define the constitutive relations for permeability, inertial resistance (Forchheimer term), and thermal dispersion specific to these bio-inspired geometries. 3. Multiscale Simulation Strategy: Utilizing the derived governing equations to perform rapid **Macro-scale CFD Simulations** of the entire Liquid-to-Liquid Heat Exchanger. This allows for predicting global performance metrics (Pressure Drop, Heat Transfer Rate) with significantly reduced computational cost while maintaining high fidelity. 4. Experimental Comparative Analysis (Validation Loop): **Prototype Fabrication:** Manufacturing the designed porous structures using **Metal 3D Printing (PBF)** with materials like Stainless Steel or Copper. **Load Bank Testing:** Conducting rigorous experiments in a controlled testbed to measure real-world thermal-hydraulic data. **Comparative Verification:** Closing the research loop by comparing mathematical Prediction, Simulation predictions with Experimental results to calibrate the coefficients of the governing equations, ensuring the model's accuracy for industrial application.
Rationale for Joint R&D Project	**1. Synergy of Theory and Practice:** This project bridges the gap between academic theory and industrial mass production. Georgia Tech provides the theoretical framework (Governing Equations), while SwirlX provides the physical realization (Manufacturing & Test Data). This framework also enables Korean SMEs to replace trial-and-error prototyping with predictive design rules, reducing development time and manufacturing risk for high-density cooling products. **2. Role of Georgia Tech (Theoretical & Computational):** Expertise in thermofluid dynamics is essential to formulate the mathematical models (correlations) for complex porous media flow, which is beyond standard industrial capabilities. **3. Role of SwirlX (Experimental & Manufacturing):** Comparative Verification: Closing the research loop by comparing **mathematical and simulation predictions** with **experimental results** to calibrate the coefficients of the governing equations, ensuring the model's accuracy for industrial application.
Key Words	1. Nature-inspired Porous Structures 2. Cooling 3. AI Data Center 4. Multiscale Thermofluid Analysis 5. Repetitive Porous Geometries

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives		1. **Derive Governing Equations:** Establish constitutive relations (friction factors, Nusselt numbers) for nature-inspired repetitive porous geometries. 2. **Develop Multiscale Simulation Model:** Create a macro-scale computational model that predicts full-scale heat exchanger performance with >90% reduced computational cost compared to DNS. 3. **Experimental Validation:** Fabricate prototypes using Metal AM (Additive Manufacturing) and acquire thermal-hydraulic data (Pressure drop, Heat transfer rate). 4. **Optimization:** Achieve a design capable of handling high heat flux through model-based optimization.			
R&D Target Specifications		**1. Model-Experiment Matching Accuracy (Primary Target):** • Deviation between Simulation prediction and Experimental data (for dP and Q) within 5~10% accuracy. **2. Robustness of Mathematical Correlations:** • Coefficient of Determination (R^2) for the derived friction factor (f) and Nusselt number (Nu) correlations: **> 0.95**. **3. Predictive Speed & Efficiency:** • Computational cost reduction compared to full-scale DNS: **> 90%** (enabling rapid design iteration).			
R&D Plan		**Year 1 (2027): Micro-scale Physics & Data Acquisition** • **[GT]** DNS analysis of cells to identify turbulence generation mechanisms. • **[SwirlX]** Fabrication of cell samples and setup of high-precision Load Bank. • **[Joint]** Derivation of initial porous media governing equations (Permeability/Inertial coefficient). **Year 2 (2028): Macro-scale Modeling & Calibration** • **[GT]** Development of macro-scale CFD solver using the derived equations.			

		• **[SwirlX]** Fabrication of mid-size prototypes via Metal 3D Printing. • **[Joint] Iterative Calibration:** Tuning equation coefficients to minimize error between simulation and experiment. **Year 3 (2029): Optimization & Blind Validation** • **[Joint] Blind Test:** Predicting performance of a *new* unseen geometry to prove predictive capability (R^2 > 0.95). • **[SwirlX]** Final fabrication of full-scale High-Flux Heat exchanger (>100 kW/rack). • **[GT]** Final report on the generalized design methodology for porous cooling structures.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Theoretical Modeling & Advanced Simulation:** • Perform high-fidelity DNS (Direct Numerical Simulation) on micro-scale unit cells. • Derive volume-averaged **Governing Equations** for momentum and energy conservation in porous media. • Develop mathematical correlations (f factors, Nu numbers) specific to the bio-inspired geometries. • Provide academic insights into turbulence suppression and heat transfer enhancement mechanisms.
Expected R&R of Korean SME Partner	**Manufacturing, Testing & Industrial Validation:** • **DfAM (Design for Additive Manufacturing):** Convert theoretical geometries into printable CAD models. • **Prototyping:** Fabricate heat exchangers using Metal 3D Printing (Stainless Steel/Copper) with high precision. • **Experimental Data Acquisition:** Conduct rigorous hydraulic and thermal performance tests (Load Bank Testing) to measure real-world Pressure Drop (dP) and Temperature (dT). • Provide feedback on manufacturability and cost constraints.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technological Achievement:** • Establishment of a validated "Digital Twin" design methodology for complex porous heat exchangers, reducing design iteration time by 50%. • Securing proprietary governing equations for high-efficiency cooling structures. **2. Commercialization Potential:** • Immediate application to SwirlX's **Vortex M1** product line for Hyperscale Data Centers (Google, AWS, Meta). • Ability to customize cooling solutions for next-gen chips (NVIDIA Blackwell/Rubin) with verified performance data. **3. Industrial Impact:** • Solving the "Thermal Wall" problem in AI data centers, contributing to energy efficiency (Lower PUE). • Strengthening the competitiveness of the Korean thermal management supply chain in the global AI infrastructure market.

6. References

(1) Schmidt, R. R., Cruz, E. E., & Iyengar, M. (2005). Challenges of data center thermal management. IBM Journal of Research and Development, 49(4.5), 709-723.

(2) Whitaker, S. (2013). The method of volume averaging (Vol. 13). Springer Science & Business Media.

(3) Vafai, K. (Ed.). (2015). Handbook of porous media. CRC Press.

(4) Adler, P. (2013). Porous media: geometry and transports. Elsevier.

(5) Yan, K., Deng, H., Xiao, Y., Wang, J., & Luo, Y. (2024). Thermo-hydraulic performance evaluation through experiment and simulation of additive manufactured Gyroid-structured heat exchanger. Applied Thermal Engineering, 241, 122402.

(6) Soulaine, C., & Tchelepi, H. A. (2016). Micro-continuum approach for pore-scale simulation of subsurface processes. Transport in porous media, 113(3), 431-456.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization